

TASMIA JANNAT

tjv7w@mst.edu — +1 573 647 2264 — LinkedIn — GitHub

EDUCATION

Ph.D. in Computer Science

Aug 2024 – Present

Missouri University of Science and Technology, Rolla, USA

Current CGPA: 4.00 [Scale 4.00]

MSc in Computer Science & Engineering

Sept 2021 - Apr 2024

Rajshahi University of Engineering & Technology (RUET)

Rajshahi, Bangladesh

CGPA: 4.00 [Scale 4.00]

Thesis Topic: Hyperspectral Image Classification Based on **Transfer Learning**.

BSc in Computer Science & Engineering

Jan 2016 - Feb 2021

Rajshahi University of Engineering & Technology (RUET)

Rajshahi, Bangladesh

CGPA: 3.72 [Scale 4.00], Class Position: 3rd [Among 131]

Thesis Topic: Dimension Reduction using PCA, LDA and t-SNE and Hyperspectral Image Classification by Using 3D **Convolutional Neural Network (CNN)**.

TECHNICAL SKILLS & RESEARCH INTERESTS

- **Research Interests:** High-Performance Computing (HPC), memory-centric and Processing-in-Memory (PIM) architectures, scalable algorithms for data-intensive workloads, graph and spatial analytics, GPU-accelerated computing, and memory-efficient LLM inference.
- **Programming:** C, C++, Python, MPI, CUDA, multithreading (C++ threads, pthreads), x86 Assembly
- **Tools and Environments:** Slurm, perf, UPMEM SDK, CUDA Toolkit, Git, VS Code, MATLAB
- **HPC Systems:** Missouri S&T Mill Cluster (CPU/GPU, Slurm) and UPMEM PIM system (multi-DPU execution, performance and communication analysis).

HPC PROJECTS

- **PIM-based R-tree on UPMEM DPUs:** Designed and implemented the first R-tree-based spatial query system on commercial UPMEM PIM DPUs for parallel batched query processing, achieving **15.23× kernel speedup** on **16M rectangles** with **3.99M queries**, and **14.54× lower energy** vs. CPU phase. (*Published at ISC High Performance 2026*)
- **PIM-based Graph Processing:** Designed **BFS-FAR** for breadth-first search on UPMEM PIM, reducing synchronization and host–DPU communication overhead across up to **2,540 DPUs**.
- **GPU R-tree Query System:** Implemented GPU accelerated range queries using a flattened BFS layout and per thread DFS traversal with pruning. Achieved about **65× speedup** over CPU on an **8.4 million** rectangle dataset with **2.4 million queries**. [Link]
- **CPU Multithreaded R-tree:** Designed a parallel R-tree search framework using STR construction, persistent thread pool, and cache-aware traversal. [Link]
- **Multithreaded Polygon Intersection:** Designed and implemented a high-performance **multi-threaded** polygon intersection project using R-tree spatial indexing and parallel query execution.

SELECTED PUBLICATIONS

Google Scholar: Profile

- **T. Jannat**, M. Gowanlock, S. Puri, “Parallel R-tree-Based Spatial Query Processing on a Commercial Processing-in-Memory System,” *ISC High Performance 2026*, Hamburg, Germany, 2026.

- **T. Jannat**, A. Subedi, S. Puri, “PolySketch: A GPU-Based Line Segment Intersection Filter for Polygon Intersection,” *Encyclopedia of GIS*, Springer, 2025.
- **T. Jannat**, M.A. Hossain, “Adapting vgg16 and resnet50 for cross-domain transfer learning on hyperspectral image,” *Proceedings of the 2024 6th ICEEICT*, 2024.

Submitted Manuscripts:

- **T. Jannat**, G. Wilcut, S. Bhowmick, S. Puri, “BFS-FAR: Fragment-Local Asynchronous Relaxation with Host Reconciliation for Breadth-First Search on UPMEM,” *Submitted to the SC26 IA³ Workshop*, 2026.
- M. Gardner, J. Harris, R. Munugala, **T. Jannat**, S. Puri, M. Gowanlock, “Large Scale Sorting on UPMEM Processing-in-Memory Architectures,” *Submitted to the SC26 Workshop on Parallel Data Storage and Workflows (PDSW)*, 2026.

EXPERIENCES

Graduate Research Assistant, Department of Computer Science Aug 2024 – Present
Missouri University of Science and Technology Rolla, USA

- Conducting NSF-supported HPC research on optimizing and accelerating data-intensive scientific workloads using Processing-in-Memory (PIM), GPU, and CPU systems, with applications to large-scale spatial and engineering datasets (see *HPC Projects* section).
- Collaborated on developing *sketch-based adaptive and hierarchical filter-refinement algorithms* for large-scale polygon intersection on GPUs.

Teaching Assistant, Department of Computer Science Fall 2025 – Spring 2026
Missouri University of Science and Technology Rolla, USA

- Introduction to Machine Learning (Fall 2025); Algorithms (Spring 2026): Grading, office hours, and student support.

Lecturer, Department of Computer Science & Engineering Dec 2021 - Aug 2024
Rajshahi University of Engineering & Technology Rajshahi, Bangladesh

- Taught undergraduate courses in parallel and distributed computing, computer networks, and programming; supervised undergrad final-year research projects in machine learning and computer vision.

Project Director of Young Professional Research Grant July 2022 - Oct 2023
Project Title: Hyperspectral Image Classification Based on 3D CNN using Mish Activation Function Rajshahi, Bangladesh

- Secured competitive research funding and independently completed the project from proposal development and implementation to analysis and peer-reviewed publication.

Funding: University Grants Commission, Bangladesh and RUET; 128,000 BDT (approximately \$1,050).

LEADERSHIP & PROFESSIONAL ACTIVITIES

- Selected as a **Student Lead Volunteer for SC26**, supporting student outreach and networking activities for the international HPC community.
- Presented research at **HPDC 2026** with a competitive student travel grant; poster presenter and student volunteer at **SC’25**.